

GNSS Reflectometry for Remote Sensing of Soil Moisture

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Abstract—Global Navigation Satellite System Reflectometry (GNSS-R) is an emerging technique for remote sensing of soil surfaces. In this work, the data collected by a GNSS receiver prototype mounted on a small aircraft are analyzed to retrieve soil moisture. Based on GNSS-R bi-static configuration, direct and reflected GPS signals were measured by a right hand (RHCP) and a left hand (LHCP) circular polarized antenna, respectively. The direct signal was used to calibrate the system. The soil moisture was retrieved from the power ratio of LHCP reflected SNR over RHCP direct SNR regardless of the surface roughness and incoherent components. A over-water calibration method was applied to improve the estimation accuracy. Results showed good correlation with the types of underlying terrain.

Index Terms—GNSS reflectometry, software defined radio, soil moisture retrieval.

I. INTRODUCTION

In Global Navigation Satellite System-Reflectometry (GNSS-R), the GPS signals reflected by the ground are measured using a ground-based or aircraft-based passive receiver. Over the last two decades, GNSS-R has gained increasing interest among the scientific community thanks to positioning systems development [1].

Soil moisture retrieval is one of the most important applications implementing GNSS-R technique. Several ground based field experiments and some airborne campaigns have shown the possibilities to retrieve soil moisture using reflected GNSS signals. Recently, there is a growing interest in applying GNSS-R techniques to signals received by UAV. But this application is limited by the hardware weight and size.

Mainly, three methods can be applied to retrieve the characteristics of the ground from the received power. In the first method, a standard ground based nearly hemispherical right hand circularly polarized (RHCP) antenna is used to receive the direct signal. The reflected signal originated from ground creates multipath effects, because it interferes with the direct signal. The total received signal amplitude has a sinusoid behavior as a function of the elevation angle. By measuring the received signal to noise ratio it is possible to retrieve soil moisture information [2], [3].

The Interference Pattern Technique (IPT) [4]-[6], is based on

the measurement of power fluctuations of the vertical polarized signal resulted from the simultaneous reception (and interference) of the direct and the reflected GNSS signals.

The bi-static method [7], [8], is based on the separate reception of direct and reflected signals with two antennas and on the separate measurement of signal powers.

These methods, however, were mainly applied on static measurements and they were seldom applied to data collected during a flight [9]-[11].

In this work, data collected by a GNSS receiver prototype developed by Istituto Superiore Mario Boella (ISMB) in the framework of the Italian project SMAT-F2 (System of Advanced monitoring of the Territory – phase 2) are analyzed. This prototype due to light weight and small size can be easily installed on small aircraft or UAV.

During the December 11th, 2014 measurement campaign, the prototype was mounted on a Digisky P92 aircraft and the area around Avigliana lakes was flew over. Both direct and reflected GPS signals were measured with a right hand circular polarized (RHCP) and a left hand circular polarized (LHCP) antenna, respectively. The reflected signal was processed with an open loop approach, in order to obtain Delay Doppler Maps (DDMs) and the corresponding Delay Waveforms. Signal to Noise Ratios (SNRs) time series were estimated from several non-coherently integrated Delay Waveforms.

Soil moisture was retrieved applying the bi-static method considering the power ratio of LHCP reflected SNR over direct RHCP SNR, regardless of the surface roughness and incoherent components. A calibration process was performed considering the signals reflected from the lakes. This light and compact GNSS receiver is a promising solution for the retrieval of soil moisture from UAV.

II. GNSS RECEIVER PROTOTYPE AND SIGNAL PROCESSING

The GNSS-R prototype mounted on a Digisky P92 aircraft is shown in Fig. 1. The hardware architecture consists of two double-chain Radio Frequency Front-Ends (FE) and a microprocessor board mounted in a carbon fiber box, specifically designed with a wing profile for aerodynamics requirements. The prototype dimensions are 75 mm x 150 mm x 250 mm and its weight is less than 3 kg making it sufficiently light and compact to be mounted on board of small aircrafts or UAV. Two antennas, one conventional GNSS L1 patch antenna pointing to the sky, and the other dual polarization (RHCP/LHCP) L1/L2 GNSS antenna pointing to the Earth surface [12] are connected to the two front-end, forming four channels, master-FE1, slave-FE1 master FE2 and slave FE2.

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The front-end receivers produced by NSL Stereo [13], fully flexible synchronized double-chain GNSS FE, are connected to a microprocessor board. The microprocessor board was developed based on the Open-Android (ODROID)-X2 platform, it is an upgrade version of ODROID-X with 1.7 GHz ARM Cortex-A9 Quad Core platform and 2GB RAM [14].

The GNSS receiver is able to work to handle signals coming from two antennas. One up-looking Right-Hand Circular Polarized antenna (RHCP), to receive the direct signal from satellites, and one down-looking antenna to measure the signals coming after reflection from ground.

The raw data were stored on board in order to be post processed. But due to the large amount of memory (GB/min) required for storing the raw data, the duration of the data collection is limited. The post-processing was made with the software SOPRANO and could work in different environment [15]. A coherent and non-coherent integration time technique was adopted during the post-processing in order to mitigate the noise. Note that the reflected signal is much weaker than the direct one and it is not continuous in time, because of the non-homogeneous characteristics of the reflected surface. Hence, in order to detect the reflected signal, a channel aiding is implemented to exploit the direct signal information.

A range of values of the delays of the reflected signals is assigned. If the reflected peak search fails, the non-coherent time is changed. The expected reflected delays were obtained depending on the direct signal delays and system geometry. A range of expected delays for processing reflected signals was defined. Therefore, the reflected delays could be synchronized with direct delays.

The optimal coherent and non-coherent time interval depend on the coherence time of the scattered signal that is determined by the receiver dynamics with respect to the scattered signal in far field [17]. In GPS-reflectometry applications, the parameters most commonly used in practice are a coherent time of 1 ms, combined with a number non-coherent sums in the order of 100 to 1000 [16]. However, for the typical altitude of the flight test analyzed in this work, the expected coherence time is 0.1-0.5 seconds.



Fig. 1. GNSS-R prototype mounted on Digisky P92 aircraft.

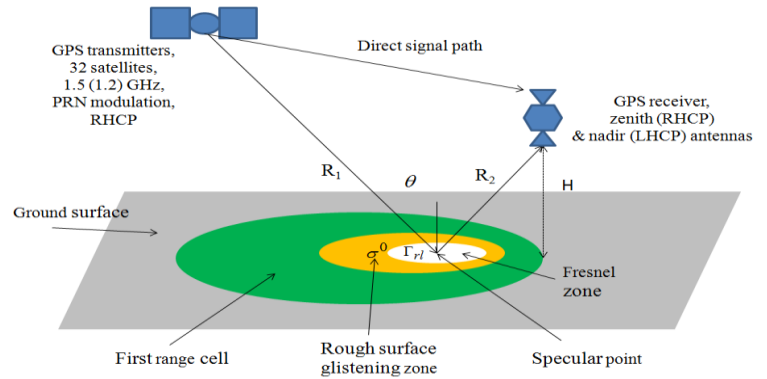
III. RETRIEVAL PROCESS FROM LHCP REFLECTED SIGNALS

In this section, assuming a perfectly smooth surface, the retrieval of the dielectric constant and of the soil moisture from reflected LHCP signals is addressed. In the case of specular reflection, the reflected GPS signals of satellites with high elevation angle (greater than 60°) are predominately LHCP component. Therefore, the dielectric constant is retrieved from the power ratio of LHCP reflected SNR over direct RHCP.

In this case, the power is reflected coherently to the receiver, and the bistatic radar equation describing the coherent component in the GPS bistatic radar (RHCP transmitted, LHCP received) is [17]:

$$P_{lr}^{coh} = R_{lr} \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 (R_1 + R_2)^2} \quad (1)$$

where P_t is transmitted signal power, G_t is the transmitter antenna gain; G_r is the receiver antenna gain and λ is the wavelength ($\lambda = 19.042$ cm for GPS L1 signal); R_{lr} is the reflectivity. Subscript lr represents the scattering when an incident signal coming from the satellite is scattered by the surface and inverts the polarization into left.



The surface reflectivity is defined as:

$$R_{lr}(\theta) = |\Gamma_{rl}(\theta)| \chi(z) \quad (2)$$

where $|\Gamma_{rl}|$ is the Fresnel reflection coefficient and $\chi(z)$ is the probability density function of the surface height z . Under the assumption of perfectly flat surface, $\chi(z)=1$.

The transmitted GPS signal is right polarized and can be write as a linear combination of horizontal and vertical polarization [18]:

$$\begin{bmatrix} \Gamma_{ll} & \Gamma_{lr} \\ \Gamma_{rl} & \Gamma_{rr} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \Gamma_h + \Gamma_v & \Gamma_h - \Gamma_v \\ \Gamma_h - \Gamma_v & \Gamma_h + \Gamma_v \end{bmatrix} \quad (3)$$

Thus, the Fresnel reflection coefficient can be written as:

$$\Gamma_{lr} = \frac{1}{2} (\Gamma_{vv} - \Gamma_{hh}) \quad (4)$$

where:

$$\begin{aligned} \Gamma_{hh}(\theta) &= \frac{\cos\theta - \sqrt{\epsilon_r - \sin^2\theta}}{\cos\theta + \sqrt{\epsilon_r - \sin^2\theta}} \\ \Gamma_{vv}(\theta) &= \frac{\epsilon_r \cos\theta - \sqrt{\epsilon_r - \sin^2\theta}}{\epsilon_r \cos\theta + \sqrt{\epsilon_r - \sin^2\theta}} \end{aligned} \quad (5)$$

being ε_r is the relative dielectric constant and θ the incident angle. Combining (1), (2) and (4), the processed SNR of peak power can be written as:

$$SNR_{peak}^{refl} = \frac{P_{lr}^{coh} G_D}{P_N} = \frac{1}{4} \frac{P_t G_t G_r \lambda^2 G_D}{(4\pi)^2 (R_1 + R_2)^2 P_N} (\Gamma_{vv} - \Gamma_{hh})^2 \quad (6)$$

where P_N is noise power and G_D is the processing gain due to the de-spread of the GPS C/A code. The SNR of peak power of direct RHCP signal is:

$$SNR_{peak}^{dir} = \frac{P_t G_t G_r \lambda^2 G_D}{(4\pi)^2 R_3^2 P_N} \quad (7)$$

where R_3 is the distance between the transmitter and receiver. It has to be noted that receiver gain G_r and noise power P_N of the direct signal (7) are not equal to those in (6) for the reflected signal. Therefore, a calibration process is needed. The ratio of the reflected SNR (6) over the direct SNR (7) is given by:

$$\frac{SNR_{peak}^{reflect}}{SNR_{peak}^{direct}} = \frac{R_3^2}{4(R_1 + R_2)^2} (\Gamma_{vv} - \Gamma_{hh})^2 C \quad (8)$$

where C is a calibration parameter summarizing the uncertainties of G_r and P_N . After over water calibration, the only unknown parameter in (8) is ε_r and it is solvable numerically.

The empirical model introduced in [19] was used to calculate the soil moisture. In this model, the general form of the polynomial for the real and imaginary part of the dielectric constant as a function of the soil moisture is written as:

$$\varepsilon = (\alpha_0 + \alpha_1 S + \alpha_2 C) + (b_0 + b_1 S + b_2 C) m_v + (c_0 + c_1 S + c_2 C) m_v^2 \quad (9)$$

where S and C are the sand and clay textural components of soil in percent by weight. Finally, soil moisture m_v can be obtained from (9) by using the coefficients for frequency values ranging from 1.4 GHz to 18 GHz reported in Table I.

TABLE I

	α_0	α_1	α_2	b_0	b_1	b_2	c_0	c_1	c_2
ε'	2.862	-0.012	0.001	3.803	0.462	-0.341	119.006	-0.500	0.633
ε''	0.356	-0.003	-0.008	5.507	0.044	-0.002	17.753	-0.313	0.206

IV. RESULTS

The GNSS receiver prototype developed by Istituto Superiore Mario Boella (ISMB) was mounted on a Digisky P92 aircraft and the area around Avigliana lakes was flew over during the measurement campaign of December 11th, 2014.

In Figure 3 the sky-plot of some of GPS satellites at the 1570th second of the flight are shown. PRN 3 (elevation 38.2°), 11 (elevation 80.3°), and 19 (elevation 46.7°) were taken into account in this analysis because the specular points corresponding to these satellites fell on the lakes surface. The measured signals were post-processed with an open loop approach, in order to obtain Delay Doppler Maps (DDMs) and the corresponding Delay Waveforms. Signal to Noise Ratios (SNRs) time series were estimated from non-coherently integrated Delay Waveforms. Both direct and reflected signals were processed to Signal to Noise Ratio and a calibration process to determine the calibration constant C of (8) and to take into account the gains of the transmitting and receiving antenna was performed through the over-water condition. The retrieval process to estimate the soil moisture from the evaluated SNR was then applied.

Figure 4 and 5 show the 40s calibrated reflectivity with respect to the specular points of PRN 3, 11 and 19, respectively. The time slots with high reflectivity are delimited by black vertical lines and they correspond to the portion of flight on the water. Using the algorithm described in section II, soil moisture corresponding to the 40s of flight was retrieved. Results are shown in Figure 6 superimposed on Google Maps. The flight route was from West to East and the receiver was mounted on the right wing. Therefore, the reflection paths were not blocked by the body of the aircraft. The moisture of the lake is shown in blue (100%) and that of the land surface in red and light red. The retrieved soil moisture shows good correlation with the type of surface.

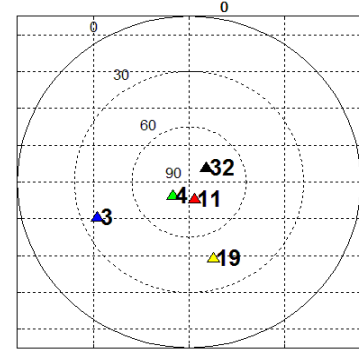


Fig. 2. Skyplot of December 11th, 2014 flight.

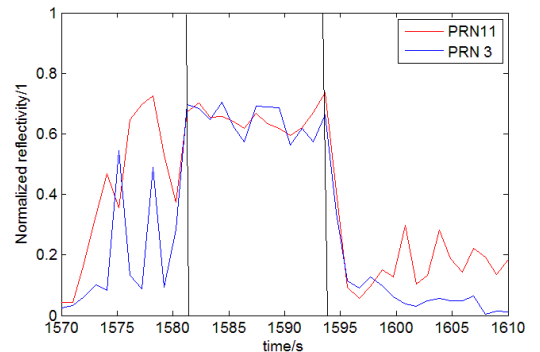


Fig. 3. Normalized reflectivity of PRN 11 and PRN 3.

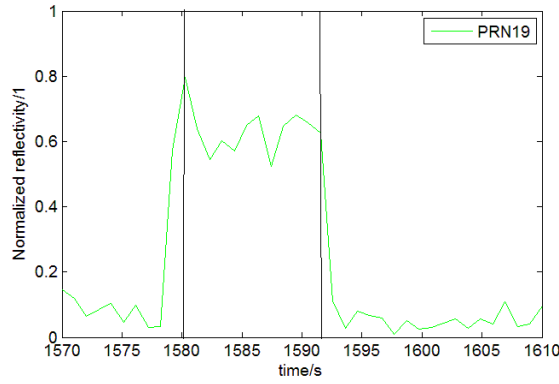


Fig. 4. Normalized reflectivity of PRN 19.

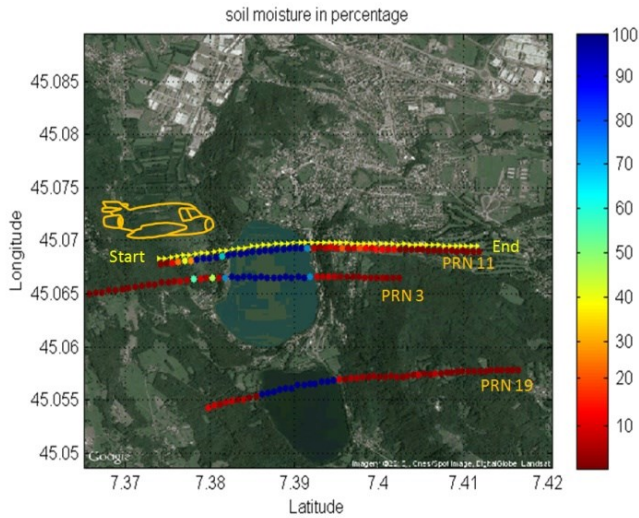


Fig. 5. Google map with the plane trajectory and the position of GPS specular reflection points.

V. CONCLUSION

Data collected during an UAV flight over the Avigliana lakes by a GNSS receiver prototype developed by the Istituto Superiore Mario Boella (ISMB) were analyzed. Due to its light weight, this receiver can be easily mounted on board of small aircraft or UAV. Both direct and reflected right-hand and left-hand circularly polarized GPS signals were measured using the corresponding up-looking and down-looking antennas. The measured signals were post-processed with an open loop approach, and Delay Doppler Maps and Delay Waveforms were retrieved. The obtained results showed good correlation with the types of underlying terrain that opens new direction in the remote sensing of natural and artificial objects.

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